

ENTITY RESOLUTION + QUALITY ASSURANCE

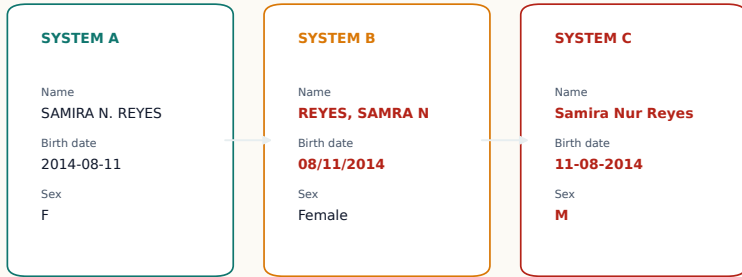
Measured Trust

An auditable path through disconnected relational data

When systems were never designed to connect, confidence must be engineered.

SYNTHETIC PROOF
CASE
school-feeding records

One person can become three records before the dashboard



Exact equality sees three records. A measured process asks whether they are one.

When records fail to connect, people disappear from the decision

01 · MISSING

The same person disappears between waves.

02 · DUPLICATE

One person is counted more than once.

03 · DISTORT

Rates and denominators quietly shift.

The interface can be correct while the identity layer is wrong.

General problem class: operational databases with no shared, reliable identifier.

A hidden answer key turns trust into something measurable

A synthetic world creates both the mess and the truth.

The generator creates spelling, date, sex-code, and identifier errors—then keeps the answer key outside the pipeline.

Stress test	Count
Name drift	396
Birth-date drift	132
Sex inconsistency	70
Missing identifier	564

Fixed-seed synthetic generator; truth artifacts remain outside the lakehouse boundary.

A trustworthy pipeline is a sequence of evidence gates



Reusable pattern: customer · patient · beneficiary · workforce · survey · education

Quality checks caught 9 in 10 failures before they reached a decision

90.75%

of deliberately planted defects were found



21

quality rules ran

1,579

known defects planted

445

extra flags for review

Synthetic proof case · fixed seed · scorecard generated by the pipeline

When a record is challenged, reviewers can see exactly why

WHERE

source file + row + field

WHEN

**pipeline run + reporting
period**

WHAT

**rule + severity + observed
value**

THEN

issue registry + review path

Exact matching left 324 of 880 known relationships unresolved

63.2%
recall

The share of known relationships recovered: 556 of 880 using exact equality alone.

Exact equality breaks when:

Names drift

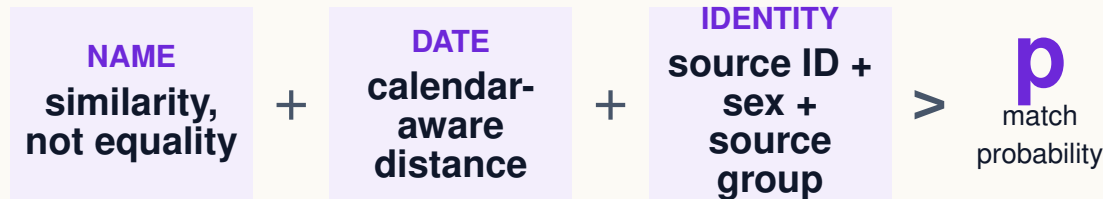
Dates wobble

Codes disagree

Precision alone cannot tell us who vanished from the join.

Deterministic baseline: 100.0% precision, 63.2% recall, 77.4% F1.

Splink learns how much each imperfect clue should count



Exact and fuzzy evidence belong inside one trained model.

Method: probabilistic entity resolution using a Fellegi–Sunter model.

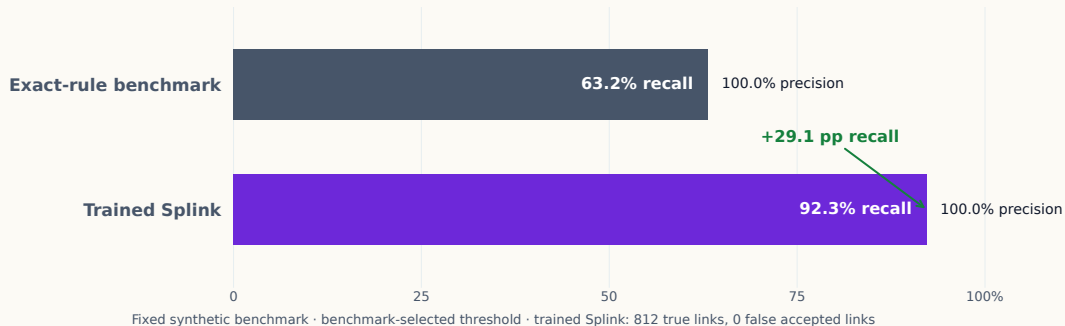
One model is trained, saved, and reloaded for inference



The weights on exact and fuzzy evidence are learned once; production links come only from the loaded model.

Config: configs/linkage_rules.yml • Runner: scripts/run_linkage_example.py

| Splink raised true links from 556 to 812



Even global Splink misses 7 of 26 cross-group moves

WHAT THIS PROVES

The full pipeline runs.

Scorecards use known truth.

Trained Splink reaches 92.3% recall with 0 false accepted links.

WHAT PRODUCTION MUST STILL PROVE

Synthetic difficulty is not field prevalence.

Seven of 26 cross-group moves remain unresolved.

Thresholds need governance and review.

Cross-group transfer recall rose from 26.9% to 73.1%—and the remaining gap stays visible.

Where databases disagree, the same trust layer can travel

CUSTOMER

**CRM + billing +
support**

BENEFICIARY

**intake + delivery
+ survey**

RESEARCH

**panel + registry +
follow-up**

PATIENT

**clinic + lab +
claims**

WORKFORCE

**HR + payroll +
training**

EDUCATION

**enrollment +
program +
outcomes**

The product is not another database; it is evidence about which connections deserve trust.

The repository carries an evidence ledger—not a promise

Assurance

Regression protection

Model contracts

Known-failure monitoring

End-to-end dependencies

Decision-ready outputs

Synthetic-only boundary

Repository evidence

407 automated tests

27 warehouse tests

21 quality rules

8 orchestrated assets

6 audience views

PASS privacy scan

Counts generated from repository artifacts at build time; privacy status reflects the synthetic-only release.

MEASURED TRUST

**A trustworthy pipeline does
not merely move data—
it measures what survives
the journey.**

Synthetic school-feeding proof case · reusable architecture for disconnected relational systems